

[09] Adjusting batch effects in microarray expression data using empirical Bayes methods

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Executive Academic Summary & Connectomic Relevance

ComBat applies empirical Bayes location (γ) and scale (δ) parameter estimation to adjust for batch effects across experimental cohorts. In our ASD research, ComBat is deployed to remove multi-site MRI scanner discrepancies across the 17 scanning centers in ABIDE I while preserving diagnostic group labels, age, and sex variance.

Mathematical & Neurobiological Foundations

- $Y_{ijg} = \alpha_g + X_{ij} \beta_g + \gamma_{ig} + \delta_{ig} \epsilon_{ijg}$
- $\gamma_{ig}^* = \frac{\tau_g^2 \hat{\gamma}_{ig} + \sigma_{ig}^2 \bar{\gamma}_i}{\tau_g^2 + \sigma_{ig}^2}$

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